The firmware works.
The upgrade doesn't.

IN THE LAB · NEW FIRMWARE

✓ **New firmware boots** Compatible

`post-quantum (ML-DSA) bootloader + image → running`

IN THE FIELD · DEVICES ALREADY SHIPPED

✕ **Update rejected** Incompatible

`deployed ECDSA bootloader rejects the ML-DSA update`

> **One failed firmware migration can turn a remote software update into hundreds of costly on-site interventions.**
>
> **ReleaseProof** finds the blocker before the rollout, with executed tests and traceable evidence.

One failed update. Thousands of devices at risk.

EXAMPLE: INDUSTRIAL SENSOR OEM — ILLUSTRATIVE SCENARIO

5,000 wireless sensors installed in plants × **5%** can't take the update and need a site visit × **€250** per technician visit

= **€62,500** 250 site visits from one blocked migration

Hypothetical inputs, not a measured ROI. Engineering rework and delay costs are not included. A design-partner pilot will measure the real numbers.

WHAT A BLOCKED MIGRATION COSTS THE MANUFACTURER

Field interventions
The fix becomes manual: a technician, a cable, a visit, device by device.

Engineering rework
The release goes back to design after it was declared ready.

Release delays
Security updates wait, and the installed fleet stays on old firmware.

Operational continuity
Sensors that feed condition monitoring and maintenance need service.

WHY NOW

SIGNING IS CHANGING

2024
NIST FIPS 204 standardises ML-DSA, a post-quantum signature for firmware signing

2026–2027
EU Cyber Resilience Act: reporting from Sep 2026, main obligations from Dec 2027

2030
CNSA 2.0 target for post-quantum firmware signing in US national security systems

Scenario inputs are hypothetical. Regulation is context, not a mandate: the CRA does not require ML-DSA, and CNSA 2.0 applies to US national security systems only. Sources: backup B3.

ReleaseProof verifies the migration before the rollout.

STEP 1

Configure

Pick the device as it is deployed today and the planned change, e.g. ECDSA → post-quantum ML-DSA.

YOU GET

A test plan for your real fleet configuration, not a generic checklist.

→

STEP 2

Test

Real bootloader builds, boot and update in emulation, with positive and negative tests.

YOU GET

Blockers found in the lab, before any device in the field is touched.

→

STEP 3

Decide

A deterministic verdict, with every finding linked to a hashed artifact.

YOU GET

A go / no-go release decision your team can review and defend.

A TYPICAL RELEASE TEST ASKS

Does the new firmware work?

✓ Compatible

RELEASEPROOF ALSO ASKS

Can devices already in the field get there?

✕ Blocker found

Most tests check the destination. ReleaseProof tests the path.

Both answers come from the same demonstrated LIVE run (wolfBoot, 87d365): candidate COMPATIBLE, transition INCOMPATIBLE. Emulation, not silicon.

From AI-assisted testing to evidence-based decisions.

AMD-HOSTED AI
Plan
Picks extra checks from a registered catalog

→

RUNNER
Execute
Real builds and emulation, every artifact hashed

→

ENGINE
Decide
Same evidence, same verdict. No LLM

→

AMD-HOSTED AI
Explain
Cites evidence IDs, proposes next steps

→

ENGINEER
Choose
Approves a proposal, which runs as a new assessment

CAPABILITY	PROCESS IMPROVEMENT	BUSINESS OUTCOME
AI-assisted test planning	Relevant checks added automatically, required ones can't be dropped	**Less manual work to prepare each validation**
Automated build + emulation	The real migration is executed on every run	**Incompatibilities found before deployment**
Evidence-based verdict	Deterministic decision, every line linked to a hashed artifact	**Traceable decisions that can be reviewed**
AI explanations and alternatives	Plain-language root cause, allowed next steps	**Faster fix, engineer stays in control**

Capabilities are demonstrated; business outcomes are expected benefits to measure in a pilot.

● EXPECTED BENEFIT

POWERED BY AMD

AMD Token Factory

Model: Google Gemma-4-31B-IT

Plans, **explains**, **proposes**: every call traced per run

No fallback to other AI providers

Builds + emulation on AMD EPYC (AWS c7a)

When evidence is missing, the answer is `INDETERMINATE`: "not proven yet", never a false green.

WORKING PROTOTYPE · DEMONSTRATED IN EMULATION

Built and tested: two real migration blockers found.

NORDIC NRF52840 + WOLFBOOT

● DEMONSTRATED

The new firmware works

...but deployed devices can't install it

- ✓ Post-quantum ML-DSA-87 bootloader and image build and boot
- ✗ Deployed ECDSA bootloader rejects the ML-DSA update
- ✗ No bootloader self-update path on this port

Candidate compatible | **Transition incompatible**

127 s	7/7	57
INCL. AMD AI	TESTS EXECUTED	HASHED ARTIFACTS

NORDIC NRF52840 + MCUBOOT (ZEPHYR)

● DEMONSTRATED

The new firmware can't be built

...and the upgrade path is blocked too

- ✗ Pinned signing tool refuses ML-DSA keys (measured by running it)
- ✗ Single-stage MCUboot can't replace itself in the field
- ✓ Tampered and wrong-key updates are rejected

Candidate incompatible | **Transition incompatible**

190 s	6/6	45
INCL. AMD AI	TESTS EXECUTED	HASHED ARTIFACTS

Same engine, two bootloader families

Runners **measure**, only the engine **decides**

Not observed = **UNKNOWN**, never a pass

LIVE runs 87d365 (wolfBoot) and 092e99 (MCUboot), 17 Oct 2026, both passing all 27 gates of the strict acceptance with Gemma on AMD. Times are for these runs only; not a performance claim.

A repeatable business for industrial IoT manufacturers.

WHO BUYS

Firmware and product-security leads

At industrial IoT OEMs, with quality and programme managers as co-sponsors.

ENTRY MARKET

Industrial sensor OEMs

Wireless condition-monitoring sensors, often hard to reach once installed.

nRF52840, our demo chip, is used in such sensors.

WHEN THEY BUY

Before a risky change

Signing-key rotation, bootloader change, new hardware revision, post-quantum roadmap.

WHY IT RECURS

Every release, not once

Each firmware, bootloader, key or platform change needs a new check, for years.

BUSINESS MODEL

● HYPOTHESIS TO VALIDATE

1 · LAND

Paid pilot

Onboard one product profile: chip + bootloader + toolchain

→

2 · RECUR

Subscription

Per product profile, with assessments on every release

→

3 · EXPAND

Release gate

CI/API integration, evidence packs, more product lines

Priced per product profile, not per device. Expansion: connected devices, critical infrastructure, long-lifecycle embedded.

WHERE IT FITS

Scanners find **what must change**. OTA platforms **deliver** the update. ReleaseProof proves **the path in between**, complementary to both.

Pilot KPIs: **blockers found before rollout** · **engineering hours saved** · **site visits avoided**

No customers, pricing or ROI are claimed. nRF52840 appears in NI MON-10411, Lynxemi and Atomation sensors (Nordic Semiconductor): hardware relevance only, not their firmware and not customers.

Make every firmware migration a decision backed by evidence.

TODAY · BUILT

Migration preflight

2 supported configurations, evidence pack, strict release-gate CLI.

NEXT · PLANNED

Closer to the field

Real dev kits, power-loss tests, STM32 and NXP, commercial bootloaders.

VISION · PLANNED

A gate on every release

A check in the release process for every device generation already shipped.

We're looking for industrial design partners

Bring one board and one planned change. We run the pilot and measure blockers found, hours saved and visits avoided.

Try it

Demo `d1yxcnwgdgjzt6.cloudfront.net`

Code `github.com/francescocarpineti091299/release-proof`

BACKUP · ARCHITECTURE AND AMD

AMD powers planning and explanations, not pass/fail decisions

Next.js UI
wizard · live run (SSE) · evidence · AMD panel

Caddy
TLS edge · /api → api · / → web

FastAPI
orchestrator · queue · SQLite · manifests

↓ internal network, no internet egress

runner-wolfboot
wolfBoot 4c79eee · wolfSSL 4aa1ad7 · Arm GCC 14.2.rel1 · Renode 1.16.1

runner-mcuboot
MCUboot 4eba8087 · Zephyr v4.2.0 · SDK 0.17.4 · imgtool 2.2.0 · Renode + own NVMC model

↓ typed observations + sha256 of every artifact

Adapters
missing observation ⇒ UNKNOWN + gap

V1.2 engine
validate · evaluate · explanation graph · canonical hash

AI gateway
planner · explainer · alternatives · assistant; schema, citation and verdict checks

↓ persisted per run

Immutable run record
verdicts · dossier · manifest.json (inputs, pins, artifact hashes, result hash) · AI traces. Runner crash ⇒ FAILED, no verdict.

INFERENCE	AMD Token Factory, OpenAI-compatible, allow-listed host
MODEL	Google Gemma-4-31B-IT
GPU	Managed, undisclosed. Labels are not attestation
FALLBACK	None. Endpoint down ⇒ AI_UNAVAILABLE, deterministic plan runs
DEMO HOST	AWS EC2 c7a.2xlarge (AMD EPYC 4th gen) behind CloudFront, per infra audit of 17 Oct 2026
SMOKE	10/10 logical calls OK with Gemma. Not a benchmark

BACKUP · DEMONSTRATED, ILLUSTRATIVE, PLANNED

Scoped engineering evidence, stated honestly

Demonstrated today

- Upstream bootloader code at pinned commits, compiled with pinned toolchains on every run
- Ephemeral signing keys; real binaries booted and updated in Renode
- Negative tests: tampered image, wrong key, old bootloader receiving the new image
- Deterministic V1.2 engine with golden-hash regression tests
- Re-evaluations as new runs with a parent reference; history immutable
- Evidence pack download and strict release-gate CLI with engine replay

Not claimed

- Results on physical silicon: this is emulation
- Power-loss / interrupted-update safety: stays UNKNOWN
- Support for arbitrary or proprietary firmware: 2 configurations only
- Certification or regulatory compliance: non-official assessment
- Semantic truth of every AI sentence: checks cover IDs and verdict labels
- Managed GPU identity, benchmarks, customers, pricing or ROI

DEMONSTRATED executed and recorded **ILLUSTRATIVE** example numbers to validate **HYPOTHESIS / PLANNED** roadmap and business model

BACKUP · LANDSCAPE, REGULATION, VALUE MEASUREMENT

Complementary to scanners and OTA, focused on the transition

Category · examples	Main documented focus	Relationship to ReleaseProof
Binarly · ONEKEY crypto inventory / PQC readiness	Binary analysis, crypto exposure; Binarly also documents reachability and CI gates	Complementary: a finding there is a natural trigger for a run here
Finite State adjacent release assurance	Evidence-linked AI, release policy gates	Closest adjacent; our focus is the deployed → candidate bootloader transition
Memfault · Mender · Foundries.io OTA / fleet lifecycle	Delivery, staged rollout, telemetry	Partners: they deliver, we pre-check the path
Renode · upstream bootloader CI building blocks	Repeatable emulation, same-scheme update tests	We add cross-generation tests, transition rules and evidence

Regulatory context, not a mandate

NIST FIPS 204 (ML-DSA), Aug 2024. EU CRA: reporting from 11 Sep 2026, main obligations 11 Dec 2027. CNSA 2.0 (US national security systems only): firmware signing exclusive by 2030. NIST IR 8547 is still a draft.

How a pilot measures value

Share of releases with a blocker found before rollout · engineering hours per migration check · site visits avoided · cost per assessed profile. Replaces the illustrative €62,500 with measured data.